

undata

A Universal Data Element Registry for Neuroscience

Building a shared language for neuroscience data schemas

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Neuroscience has
no shared language
for data elements.

5 major ecosystems × 5 different formats × 0 interoperability

BIDS • NWB • DANDI • openMINDS • AIND

Three Problems Nobody Has Solved

1 SCHEMA FRAGMENTATION

Same concept, different formats

JSON Schema, YAML, LinkML, Python, JSON-LD

No way to ask: "What does BIDS call the thing NWB calls ElectrodeArray?"

2 IDENTITY CONFLATION

What something IS ≠ where it came from
'age' in BIDS and 'age' in AIND are treated as different things despite being semantically identical

3 INVISIBLE TRANSFORMS

Silent data transformations

Unit conversions, type coercions

happen in ad-hoc scripts

Breaking reproducibility

These problems scale with every new dataset, standard, and ecosystem.

Why This Matters

- Researchers waste weeks mapping between formats manually
- Cross-ecosystem analyses require heroic bespoke engineering
- Transformations are undocumented □ results not reproducible
- New standards fragment the landscape further
- No machine-readable crosswalk between any two ecosystems

The cost compounds with every dataset, every study, every collaboration.

Three Core Innovations

CONTENT-ADDRESSED IDENTITY

SHA-256 hash from semantic properties
(data type, unit, pattern,
constraints,
ontology grounding)

Two elements from different sources
describing the same concept □
same hash automatically

IDENTITY ≠ PROVENANCE

An element's identity (what it IS)
is separate from provenance
(where it came FROM)

A single element accumulates
provenance from multiple sources
instead of duplicating

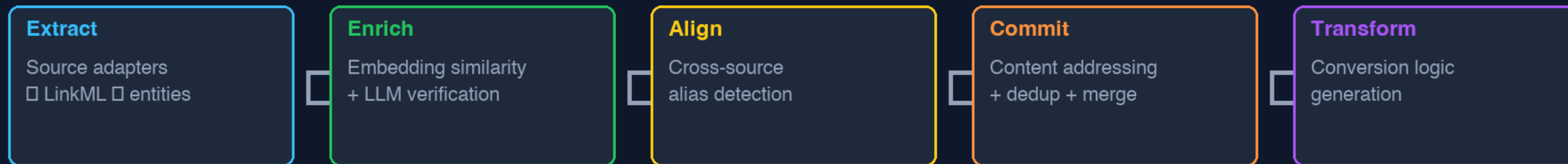
EXPLICIT TRANSFORMS

Every conversion documented:

- Unit conversion (years □ months)
- Type coercion (float □ string)
- Value mapping (M/F □ male/female)

Content-addressed with their own
provenance chains

The undata Pipeline



Sources:

- BIDS (JSON Schema)
- AIND (JSON Schema)
- NWB (Python/hdmf)
- OpenNeuro (TSV/BIDS)
- DANDI (Pydantic)
- ReproSchema (JSON-LD)
- openMINDS (JSON-LD)
- NDA (REST API)

Knowledge Service:

NCIT (209K) • PATO (2.8K) • NCBITaxon (2.7M) • UBERON • HP • DICOM (5K) • RadLex (46K) • NIDM • EDAM

13 ontologies • 2.99M terms • 268K embedded • pyoxigraph RDF store

Three-Layer Architecture

FRONTEND — Next.js

Apollo Client + Tailwind CSS
Browse, search, curation, chat
Entity navigator (ER diagram)
Infinite scroll + server-side sort
Real-time LLM curation assistant

BACKEND — FastAPI

Strawberry GraphQL + PostgreSQL 16
pgvector for embeddings
JSONB for flexible entity storage
Keycloak OIDC authentication
PROV-O audit trail for all mutations

LIBRARY — Pure Python

No DB dependencies, standalone CLI
All pipeline logic: extract → transform
StorageBackend protocol (file or DB)
436 tests, Python 3.14
uv for dependency management

StorageBackend Protocol:

read() • write() • list() • exists() • delete() • merge_provenance() • count()

Two implementations: FileBackend (YAML + Parquet + pyoxigraph) | DatabaseBackend (PostgreSQL + pgvector)

Content-Addressed Entity Model

Element

Field-level definition
age, weight, diagnosis
type, unit, constraints

- sha256
- semantic{}
- provenance[]
- annotations[]

Schema

Class-level definition
Subject, Session, Electrode
properties, inheritance

- sha256
- properties[]
- subclass_of
- provenance[]

Value

Categorical concept
male, right-handed, MRI
ontology grounding

- sha256
- label
- value_type
- provenance[]

ValueSet

Named collection
handedness_options
modality_types

- sha256
- name
- members[]
- provenance[]

+ Transform (conversion between elements) + CurationFlag (quality review) + AuditLog (provenance trail)

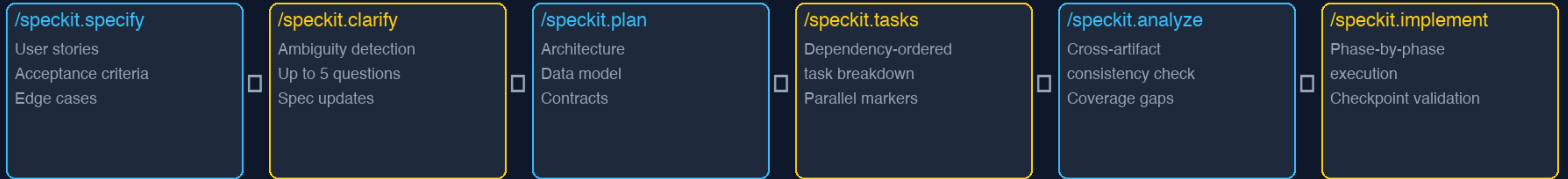
All entities: staged ☐ curated through curator review. No automated changes without evidence-based review.

Multi-Precision Enrichment

- Embedding similarity: all-MiniLM-L6-v2 $\square$ cosine similarity to 268K ontology terms
- SKOS mapping relations: exactMatch (≥ 0.95) • closeMatch (≥ 0.85) • broadMatch • relatedMatch
- LLM verification for borderline matches (0.5–0.7 score) via litellm
- Ontology hierarchy traversal: ancestors added as broadMatch (multi-precision)
- Evidence Chain for every annotation: similarity score + URI verification + reasoning
- Curated annotations protected from re-enrichment

NCBITaxon filtered to 20 neuroscience-relevant species (from 2.7M terms)

Engineering Process: Speckit Lifecycle



38 features specified and implemented through this lifecycle

Project constitution (v2.1.0) with 7 core principles enforced at every stage

Project Constitution: 7 Core Principles

I. Simplicity First

YAGNI at all levels — complexity must be justified

II. Test-Driven Development

Red-Green-Refactor, non-negotiable

III. API-First Design

Contracts before implementation

IV. Observability

Structured JSON logs, no silent failures

V. No Deprecation

System has no consumers — rewrite freely

VI. Environment Isolation

uv + venvs, bridge venv exception for adapters

VII. Developer Experience

docker compose up □ everything works

38 Features: From Brainstorm to Production

Iteration 1: Exploration (001-027)

- 001-010: Schema integration, backend, explorer, migration, enrichment
- 011-020: Metamodel, provenance, fullstack, ingestion overhaul, extraction
- 021-027: Source acquisition, repo restructure, ontology, staged enrichment

Iteration 2: Production (028-038)

- 028-029: Storage abstraction, backend service layer
- 030-032: Frontend integration, CivicDB-inspired UI, authentication
- 033-034: Unit standardization, curation interface
- 035-037: UX overhaul, knowledge service, data export
- 038: System hardening — 47 tasks, 9 user stories

The Registry: Scale & Coverage

7,055

Elements

1,005

Schemas

15,881

Values

2,426

Value Sets

15

Transforms

7 Sources:

BIDS

585 elements • JSON Schema

NWB

179 elements • Python/hdmf

DANDI

398 elements • Pydantic

openMINDS

473 elements • JSON-LD

AIND

556 elements • JSON Schema

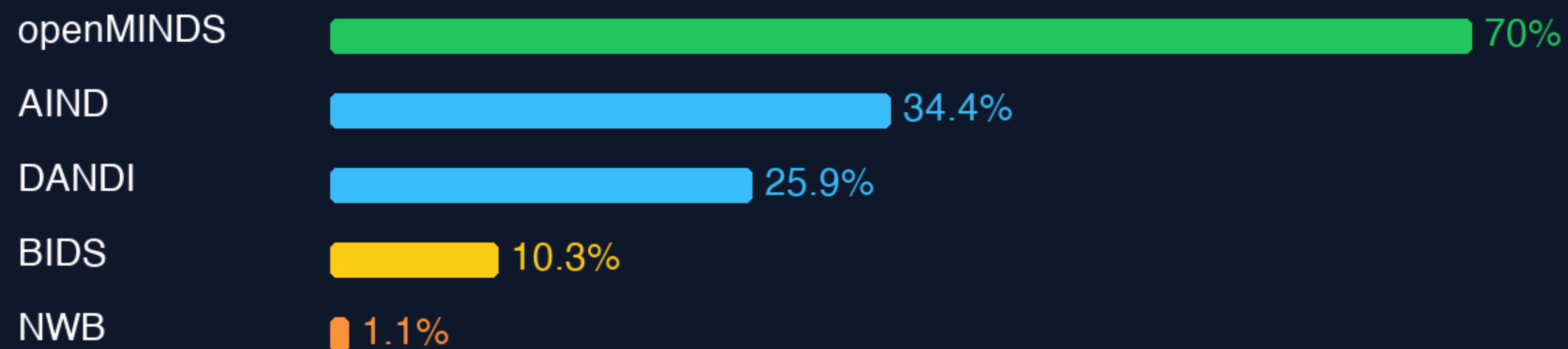
OpenNeuro

28 elements • TSV/datalad

ReproSchema

4,836 elements • JSON-LD

Enrichment Coverage



13 Ontologies • 2.99M total terms • 268K embedded

- Evidence Chain: similarity score + URI verification + reasoning chain
- LLM verification (gpt-5.4-nano: 0.06s/pair, ollama/qwen3.5: 0.5s/pair)
- NCBITaxon filtered to 20 neuroscience species (from 2.7M terms)
- Curated annotations protected from re-enrichment

Knowledge Service: Expanding the Semantic Universe

ONTOLOGIES

Core: NCIT, PATO, UBERON, HP, OBI
Neuro: NIDM, CogPO, NIF-std, ReproNim
Imaging: DICOM, RadLex, NeuroNames
New: HoMBA (brain anatomy)
pyoxigraph RDF store with SPARQL
Checksum-based dedup per source

DATA REPOSITORIES

OpenNeuro (BIDS + metadata, datalad)
DANDI (NWB + schema.org)
openMINDS instances (4,390 entries)
NDA data dictionary API
ReproSchema library (JSON-LD)
Adapter pattern: BaseAdapter +
registry

KNOWLEDGE BASES

Cognitive Atlas (concepts □ tasks)
InterLex (NIF registry)
NITRC (tools + capabilities)
NIH CDE Repository
CDISC, FHIR, schema.org
Discovery service polls APIs daily

Adapter Architecture: 8 Source Adapters

BIDS	LinkML-first, raw JSON Schema descriptors
NWB	Python/hdmf introspection via bridge venv (Python 3.12)
DANDI	Pydantic model introspection, inheritance handling
openMINDS	JSON-LD □ LinkML, 4,390 instance values
AIND	JSON Schema with \$ref resolution, enum extraction
OpenNeuro	datalad Python API, TSV/CSV + JSON sidecar scanning
ReproSchema	JSON-LD with relative path ref resolution
NDA	REST API, data dictionary extraction with value ranges

All adapters: BaseAdapter protocol □ `extract()` □ `list[ClassifiedEntity]`

Entity types: **CLASS**, **ATTRIBUTE**, **ENUM_VALUE**, **VALUESET**

Content-Addressed Identity: How It Works

TWO-MODE HASHING

Mode 1 — Ontology-Anchored:

hash(data_type + unit + pattern + constraints
+ primary_ontology_uri)

□ Same concept across sources gets same hash

Mode 2 — Structural Fallback:

hash(data_type + unit + pattern + constraints
+ class + attribute + description)

□ Unique within context when no ontology match

Short key: 12 hex chars from SHA-256

Sufficient for current scale (7K+ elements)

PROVENANCE ACCUMULATION

Element: age (sha256: 240539a4c71f...)

Provenance[0]: source=bids, class=participants

Provenance[1]: source=nwb, class=Subject

Provenance[2]: source=aind, class=Subject

Provenance[3]: source=openneuro/ds000228

One entity, many origins

Cross-referenced automatically

Browsable ER-diagram navigator

Transform Generation: Three Strategies

1. SHARED ONTOLOGY URI

Elements with same primary annotation
e.g., both mapped to NCIT:C25150 (Age)
Original method: 15 transforms generated
Detects: identity, unit_conversion,
type_conversion, value_mapping,
structural, unknown

2. NAME-BASED MATCHING

Group by provenance name (case-insensitive)
Cross-source only (BIDS age $\square$ NWB age)
Type compatibility check
Dramatically increases coverage:
100+ transforms (up from 15)

3. EMBEDDING SIMILARITY

Cosine similarity between element
description embeddings
Threshold: 0.8 (cross-source only)
Catches semantic matches missed by
exact name or ontology matching
Capped at 500 pairs for efficiency

Many-to-one support: source_elements[] field for composite transforms (age + age_unit $\square$ age_in_years)

Evidence-Based Curation Workflow

EVIDENCE CHAIN (Every Proposal)

similarity_score: 0.85
similarity_method: cosine_embedding
source_text: 'Age of participant in years'
target_term_uri: NCIT:C25150
target_term_label: 'Age'
target_term_definition: '...'
uri_verified: true ☐
reasoning: 'Cosine similarity 0.85...'
No hallucinated confidence.

Every claim backed by verifiable evidence.

CURATION ROLES

Contributor (default):

- Suggest annotations, flag issues

Curator (reviews):

- Approve/reject proposals
- Resolve curation flags
- LLM-assisted chat curation

Admin (manages):

- Import registries, manage ontologies
- Version dependency management

AUDIT TRAIL

PROV-O style for every mutation:

- agent (who)
- activity (what)
- entity (on what)
- generated_entity (new version)
- timestamp

Queryable by entity, user, time

14,984 curation flags tracked

All mutation types recorded

LLM-Powered Curation Assistant

- Real-time chat connected to configurable LLM backend (litellm: OpenAI, ollama, Anthropic)
- Entity context injected into every conversation (all fields, provenance, annotations)
- Auto-suggest improvements on entity load (missing annotations, unit inference, description quality)
- Tool execution: `lookup_ontology_term`, `propose_entity_change`, `fetch_entity`, `trigger_ingestion`
- Proposals appear as reviewable diffs with evidence chains in right panel
- Borderline candidates (0.5-0.7 score) batch-verified via LLM with cache

[SSE streaming](#) • [Tool execution](#) • [Evidence chain display](#) • [Diff review](#)

Search: Lexical, Semantic, and Combined

LEXICAL SEARCH

PostgreSQL tsvector full-text search
ILIKE fallback for simple queries
Cross-entity: elements, schemas,
values, valuesets
Exact keyword matching

SEMANTIC SEARCH

pgvector nearest-neighbor search
Query $\square$ embedding $\square$ cosine distance
Finds conceptually related terms:
"brain area" $\square$ brain_region,
cortical_area, anatomical_region

COMBINED MODE

Default: both lexical + semantic
Lexical matches first (exact)
Semantic matches after (related)
Similarity scores displayed
Mode toggle in search page UI

UI: Entity Navigator — ER Diagram in a Browser

- Every entity links to related entities — schemas □ elements □ values □ transforms
- Server-side sorting on all columns (not client-side re-sort of loaded page)
- Infinite scroll with IntersectionObserver (sentinel 200px before end)
- Source filtering on every browse page (BIDS, NWB, DANDI, openMINDS, AIND, ...)
- Cross-entity search with results grouped by type
- Annotation chips with expandable evidence chains
- Provenance badge strips showing all source attestations
- Compact property tables resolving sha256 □ EntityTag chips

UI: Entity Detail — Full Semantic Context

- Identity block: sha256, data type, unit, unit URI, pattern, min/max, value domain
- Semantic fields: description, question text, response options
- Ontology annotations with SKOS relation, score, evidence chain
- Provenance entries with W3C PROV-O metadata (source, class, activity, timestamp)
- Cross-references: schemas containing this element, transforms involving it
- Curation flags with context, LLM verification, resolve buttons
- Versioning: superseded_by link chain, curation_update transforms

Feature 038: System Hardening

47 tasks • 9 user stories • 12 phases • 47/47 complete

US1 P1	LLM Curation Chat	Auto-suggest, evidence chains, tool execution
US2 P1	Name-Based Transforms	100+ transforms (up from 15), many-to-one
US3 P1	Additional Sources	OpenNeuro, ReproSchema, NDA adapters
US4 P2	Search Modes	Lexical/semantic/both with pgvector
US5 P2	Ontology Admin	pyoxigraph store info, NCBITaxon filter
US6 P2	Server-Side Sorting	All browse pages, infinite scroll
US7 P3	Audit + Downloads	PROV-O audit, nightly exports
US8 P3	CI + Pipeline	Action v6, index auto-rebuild, LLM enrichment
US9 P1	Versioned Dependencies	Checksum detection, auto re-enrich

Evidence Chain: No Hallucinated Confidence

Every automated proposal carries verifiable evidence:

EVIDENCE CHAIN STRUCTURE

Similarity Score: 0.85 [85%]	
Method: cosine_embedding	
Source: 'Age of participant (years)'	
Target: NCIT:C25150 'Age'	
Def: 'How long something existed'	
URI Verified: ☐ (HTTP 200)	
Reasoning: Cosine similarity 0.85	
between element description and	
term label+definition. Relation:	
skos:closeMatch	

THREE VERIFICATION LAYERS

1. SEMANTIC SIMILARITY
Embedding cosine score (0.0-1.0)
Threshold: 0.7 for annotation
0.95+ for auto-assign
2. LINK VERIFICATION
HTTP HEAD check on proposed URI
Detects stale/broken ontology refs
3. REASONING CHAIN
Step-by-step explanation
LLM-generated for borderline matches
Template-generated for embedding matches

Versioned Dependency Management

- Scheduled checksum comparison for all registered ontologies and sources
- When change detected □ automatic re-enrichment of affected entities
- Curator-approved annotations preserved (curated_annotations field)
- Only automated annotations from old version are re-evaluated
- VersionTransition recorded in provenance: old checksum □ new checksum □ timestamp
- checkDependencyVersions GraphQL mutation for manual trigger
- HoMBA ontology loaded from brain-bican GitHub releases with OWL□TTL fallback

Polyglot Storage: Right Tool for Each Data Type

Data	Pattern	File Backend	DB Backend
Entities	CRUD, filter, paginate	YAML	PostgreSQL JSONB
Ontology terms	Graph traversal, SPARQL	pyoxigraph	RDF store
Embeddings	Nearest-neighbor, cosine	Parquet + numpy	pgvector
Full-text search	Keyword, faceted	grep/memory	tsvector
LLM cache	Key-value lookup	JSON file	PostgreSQL
Curation flags	CRUD, status filter	YAML	PostgreSQL
Audit log	Append-only, time-range	YAML	PostgreSQL

GraphQL API: Single Unified Interface

QUERIES

element(sha256) □ Element
browseElements(source, sort, search)
browseSchemas / browseValues /
browseValuesets
search(query, mode) □ SearchResult[]
curationQueue(flagType, status)
flagsForEntity(entityType, entityRef)
schemasUsingElement(sha256)
transformsForElement(sha256)
ontologyStoreInfo □ OntologyStoreEntry[]
auditLog(entity, agent, activity)
enrichmentProposals(entity, status)
releases(releaseType)

MUTATIONS

resolveFlag / batchResolveFlags
updateElement / updateSchema /
updateValue
approveAnnotation / rejectAnnotation
versionElement(sha256, changes)
approveIngestion / rejectIngestion
reviewProposal(id, decision)
requestEnrichment(entity)
importRegistry(path)
exportRegistry(version)
checkDependencyVersions
tagRelease(version)

Features:

- Relay-style cursor pagination
- Keycloak OIDC auth (3 roles)
- PROV-O audit on every mutation
- SearchMode enum (lexical/semantic/both)
- Server-side sort (sortBy/sortOrder)
- Apollo paginationMerge (dedup)

Testing: 436 Library Tests + Full Stack CI

436

Library Tests

16

Transform Tests

9

CI Workflows

0

TS Errors

- pytest + pytest-asyncio for all Python tests
- TypeScript strict mode with zero non-test errors
- GitHub Actions: library, backend, frontend, e2e, lint, build-images
- Actions updated to v5/v6 (Node.js 24 compatible)
- Docker compose for integration testing
- Property-based tests for content addressing

Standing on the Shoulders of Giants

NEUROSCIENCE STANDARDS

BIDS (Brain Imaging Data Structure)

Gorgolewski et al., 2016

NWB (Neurodata Without Borders)

Teeters et al., 2015

DANDI (Distributed Archives for Neuro)

Halchenko et al., 2024

openMINDS (EBRAINS metadata)

Zehl et al., 2023

AIND (Allen Institute)

Allen Institute, 2023

KNOWLEDGE RESOURCES

NCIT — NCI Thesaurus (209K terms)

UBERON — Anatomy ontology

PATO — Phenotype/trait ontology

RadLex — Radiology lexicon (46K)

DICOM — Medical imaging (5K)

NIDM — Neuroimaging data model

HoMBA — Brain anatomy (brain-bican)

Cognitive Atlas — Poldrack et al.

ReproSchema — ReproNim project

NDA — NIMH Data Archive

TECHNOLOGIES

LinkML — Moxon et al., 2021

SKOS — W3C vocabulary mapping

W3C PROV-O — provenance model

sentence-transformers (all-
MiniLM-L6-v2)

pyoxigraph — RDF store

pgvector — PostgreSQL embeddings

litellm — LLM abstraction

datalad — distributed data mgmt

CivicDB — community curation model

QUDT — unit vocabulary

Community Curation Model (Inspired by CivicDB)

- Three-tier graduated engagement: Contributor □ Curator □ Admin
- Revision-based workflow: suggestions with evidence, curator approves/rejects
- Polymorphic concerns: any entity type can be commented, flagged, subscribed to
- Evidence panels: automated match candidates + scores, LLM verification, related entities
- Activity feed filterable by action, user, source
- Organization-level attribution and contribution statistics
- Notification system for subscribed entities and flags

Every change goes through curation — no automated change applied without review.

Data Export & Downloads

- Nightly export scheduler: asyncio background task, daily archive production
- Export format: YAML entities + Parquet embeddings + manifest
- Release records: version, file size, entity counts, download count
- Static file serving at `/api/downloads/`
- Downloads page in UI with version, date, size, entity counts
- Versioned releases: tag any nightly export with a version string

Cross-Ecosystem Impact

Before undata:

- Each ecosystem defines its own 'age' field independently
- Converting between BIDS and NWB requires custom scripts per field
- No machine-readable way to discover equivalent concepts
- Transformation provenance lost in pipeline scripts

After undata:

- One 'age' element (sha256: 240539a4c71f...) with provenance from 5+ sources
- Transforms auto-generated with documented conversion logic
- Semantic search finds related concepts across all ecosystems
- Every annotation backed by evidence chain and curation review

Reproducibility Through Provenance

- Content-addressed identity: same semantic content $\square$ same hash, deterministic
- W3C PROV-O audit trail: who changed what, when, why, and what was generated
- Version transitions recorded: old ontology $\square$ new ontology $\square$ affected entities
- Transform provenance: explicit conversion logic, not hidden scripts
- Curation flags: machine-generated quality concerns requiring human review
- Nightly exports: timestamped, versioned snapshots of the complete registry

Scalability & Performance

<5s

Chat response

<1s

Search latency

<10m

Pipeline run

7K+

Elements indexed

- PostgreSQL 16 JSONB for flexible entity storage at scale
- pgvector for embedding nearest-neighbor in milliseconds
- Cursor-based pagination (not offset) for consistent performance
- Ontology store handles 2.99M terms in pyoxigraph without issues
- Embedding index: 268K terms, NCBITaxon filtered to avoid bloat
- Apollo cache with paginationMerge for frontend efficiency

Developer Experience: One Command to Everything

\$ docker compose up -d

☐ Database seeded • Backend running • Frontend live • Auth configured

- Library standalone: `uv run undata-library pipeline --source bids` (no Docker needed)
- Hot reload for backend and frontend development
- `docker-compose.override.yml` for full registry mounting (not committed)
- Curated seed subset (70 elements, 7 sources) committed for quick start
- Full registry at `~/.cache/undata/registry/` for comprehensive testing
- Ontology store mounted from host for `pyoxigraph` access

What's Next

NEAR TERM

- Run full enrichment with all 7 sources
- Generate 100+ transforms with all 3 strategies
- Deploy to staging environment
- Community beta with initial curators
- Load HoMBA and additional ontologies
- OpenNeuro batch ingestion (100+ datasets)

MEDIUM TERM

- Custom LLM fine-tuning for domain
- Real-time collaborative curation
- Cross-registry federation
- API consumers and integrations
- Automated pipeline scheduling
- Stats/mapping repository ingestion

LONG TERM

- Community-driven ontology creation
- Federated registry network
- Standard adoption by ecosystems
- Training data for neuroscience NLP
- Integration with analysis pipelines

Key Takeaways

- Content-addressed identity solves schema fragmentation across 7+ neuroscience ecosystems
- Evidence-based curation prevents hallucinated confidence — every claim verifiable
- 38 features, 436 tests, 7,055 elements from a rigorous speckit-driven process
- Three enrichment strategies: embedding similarity + LLM verification + ontology hierarchy
- Three transform strategies: shared URI + name matching + embedding similarity
- Full audit trail with PROV-O provenance for every mutation
- Library + backend + frontend — all open, all tested, all documented

By The Numbers

38

Features specified

47

Tasks in hardening

436

Library tests

7,055

Elements

15,881

Values

1,005

Schemas

2,426

Value Sets

9

Ontologies loaded

268K

Terms embedded

8

Source adapters

7

Constitution principles

3

Architecture layers

Thank You

undata: A Universal Data Element Registry
for Neuroscience

github.com/sensein/undata

satra@mit.edu

Built with: Python 3.14 • FastAPI • Next.js • PostgreSQL • pyoxigraph

Powered by: speckit lifecycle • litellm • sentence-transformers • datalad